

LUKASS KELLIJS

Yale University, New Haven, CT



About

I am an undergraduate student at Yale University (but am originally from Riga, Latvia). My main interests relate to physics and engineering, and their intersection with machine learning (I'm excited about new ways to create). I have personal interests in space and climate sciences. I am involved in and organize various STEM-related projects nationally and internationally.

Education

Yale University

September 2023 – May 2027 (New Haven, USA)

Applied Physics (B.S.) '27 · GPA: 3.98/4.00

Engineering High School of Riga Technical University

September 2020 – July 2023 (Riga, Latvia)

Research

Research Engineer Intern, HUD AI

May 2026 – August 2026 (San Francisco, USA)

Research engineering on RL environments and quality assessment for robotics models and robotics data.

Logan Wright Applied Physics Laboratory, Yale University

September 2024 – Present (New Haven, USA)

Using Machine Learning methods for the inverse design of optical computing systems.

Computational and Applied Mathematics Laboratory, ETH Zürich

May 2025 – July 2025 (Zürich, Switzerland)

Summer research in scientific machine learning: physics-informed losses for training neural operator models.

Activities

Co-President at Yale AI Association

September 2025 – Present (New Haven, USA)

Founded club of more than 100 members, propose and organize events, lead club work.

Head / Public Relations Manager at European Space Camp (ESC)

September 2022 – September 2025 (Andøya, Norway)

Largest educational space camp in Europe. Led a team of 6 international volunteers for the past 3 years: planning, organizing, and promoting ESC.

Member at Yale Undergraduate Aerospace Association

September 2023 – September 2025 (New Haven, USA)

Board member, led projects in CubeSat (Mechanical) and Liquid Rocket (Propulsion) teams.

Avionics Team Lead at Riga Technical University High Power Rocketry Team

October 2020 – September 2023 (Riga, Latvia)

First rocketry team in Latvia. Led the Avionics sub-team, developing and testing all launch electronics.

Achievements

International Physics Olympiad 2022, 2023

Bronze Medal

European Physics Olympiad 2022

Bronze Medal

Latvian National Physics Olympiad 2021, 2022, 2023

1st place

EU Contest for Young Scientists 2022

Natural Biodiversity Award

Relevant Coursework

Graduate

S&DS 6890 Scientific Machine Learning
ENV 5940 Global Carbon Cycle (Audit)

CPSC 5710 Trustworthy Deep Learning

Undergraduate

CPSC 4520 Deep Learning Theory and Applications
PHYS 4500 Thermodynamics and Statistical Mechanics
PHYS 4300 Electromagnetic Fields and Optics
ENAS 4041 Applied Numerical Methods for Differential Equations

PHYS 4400 Quantum Mechanics I and II
S&DS 3510 Stochastic Processes
AMTH 3640 Information Theory
CPSC 4870 3D Spatial Modeling and Computing

Current

APHY 5120 AI for Applied Physics Research
APHY 4480 Solid State Physics I

CPSC 4420 Theory of Computation

Links

LinkedIn
GitHub
Portfolio

Contact

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Riga, Latvia / New Haven, USA

Skills

Python, ML (PyTorch), Data Analysis (NumPy, SciPy, Pandas), C++, Matlab, R, JavaScript, Electronics, Embedded Programming, CAD (Solidworks, Onshape), 3D Printing, Graphic Design, LaTeX

Interests

Deep Learning, Reinforcement Learning, Physics, Engineering, Inverse Design, Sensor Technologies, Robotics, Space, Earth and Planetary Sciences, Climate Sciences, Green Energy

Languages

Latvian, English, German, French

Hobbies

Climbing, Basketball, Skiing, Snowboarding, Downhill Longboarding, Guitar, Reading